Computer Graphics (CS7.302)

Line and Polygon Clipping

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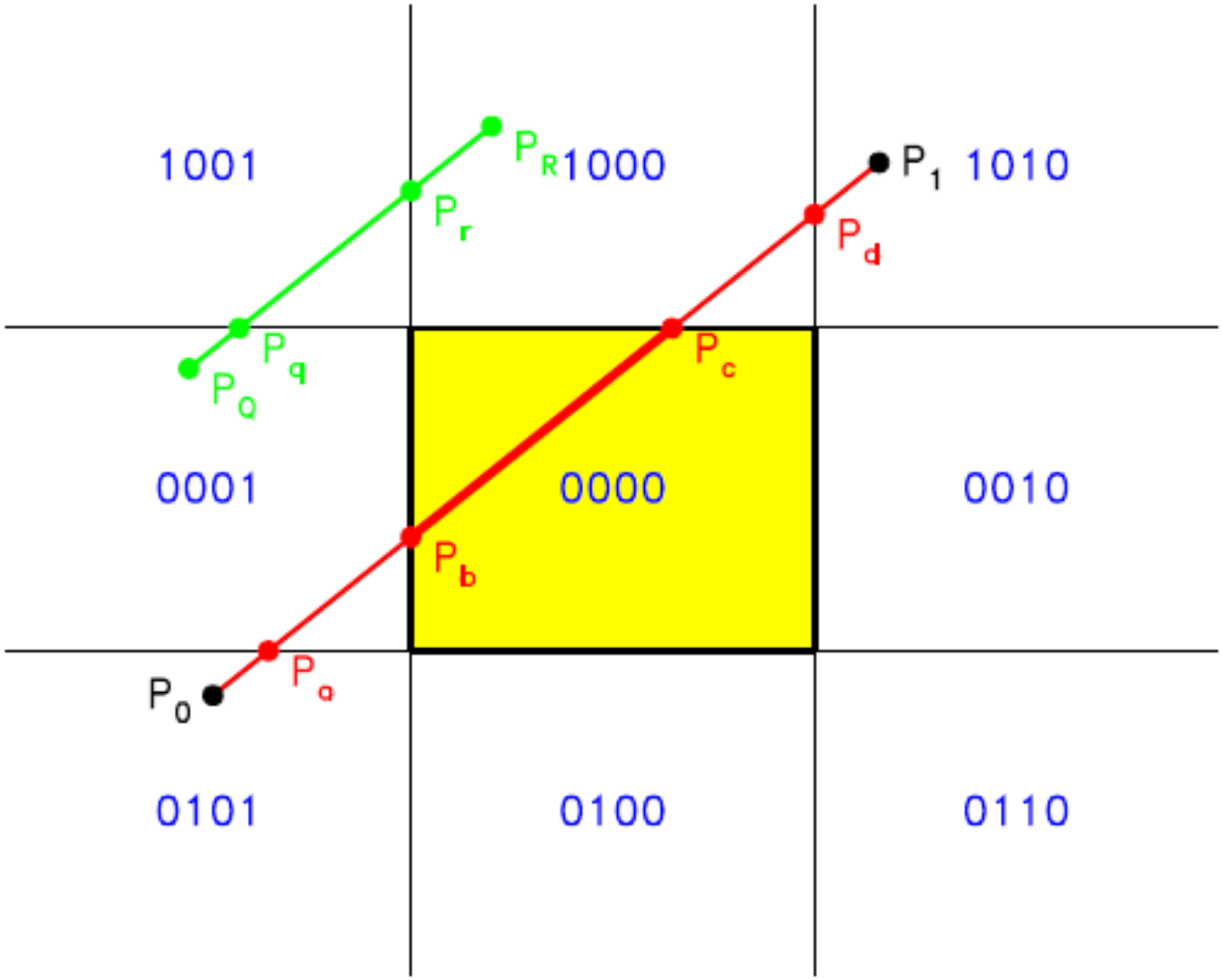
Clipping

Line Clipping

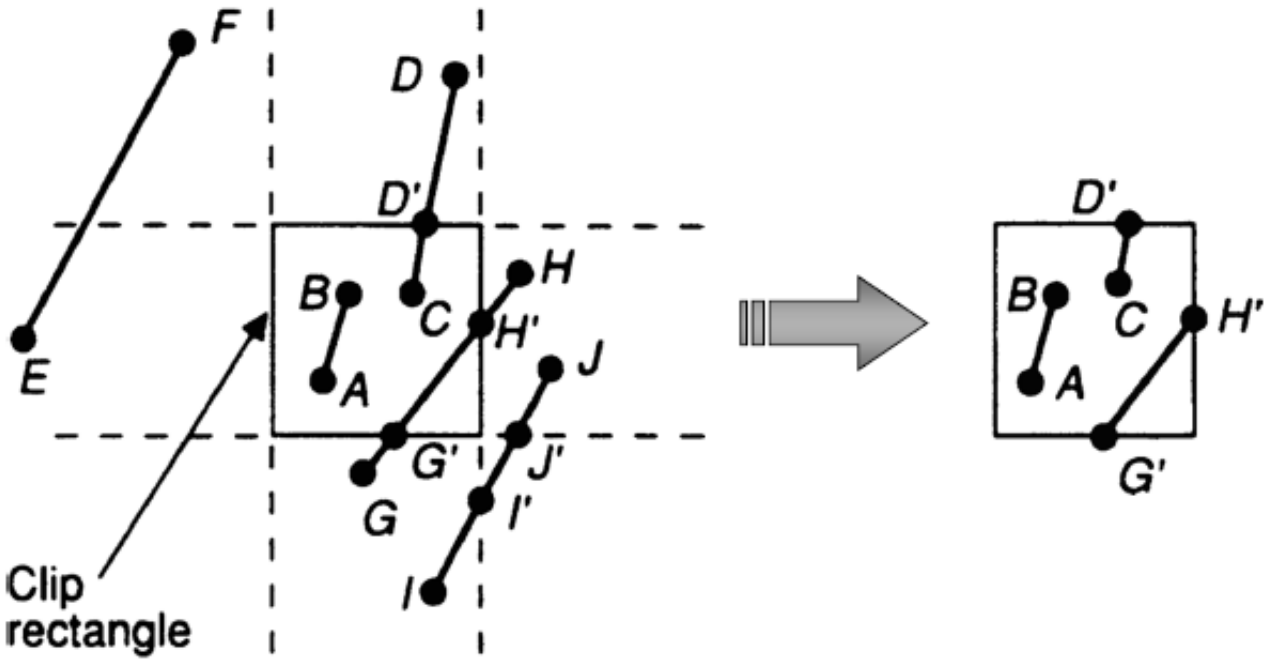
Algorithms

- Brute force
- Cohen Sutherland
- Liang Barsky

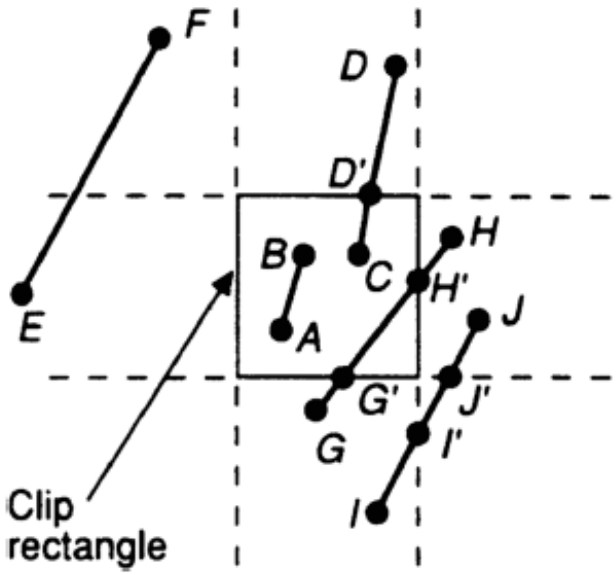
Cohen Sutherland



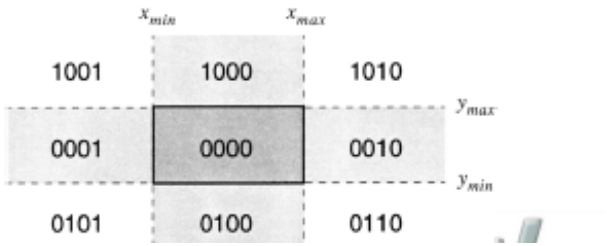
Cohen Sutherland



Cohen Sutherland



Line	Code 1	Code 2	C1 ∧ C2	C1 ∨ C2
A⇒B	0000	0000	0000	0000
C⇒D	0000	1000	0000	1000
E⇒F	0001	1001	0001	1001
G⇒H	0100	0010	0000	0110
I⇒J	0100	0010	0000	0110



- Trivial Accept $C_1|C_2 = 0$
- Trivial Reject $C_1 \& C_2 \neq 0$

Cohen Sutherland

- Define clip code of a point as four bit value TBRL, where

$$T = \begin{cases} 1, y > y_{max} \\ 0, y \leq y_{max} \end{cases} \quad B = \begin{cases} 1, y < y_{min} \\ 0, y \geq y_{min} \end{cases}$$

$$R = \begin{cases} 1, x > x_{max} \\ 0, x \leq x_{max} \end{cases} \quad L = \begin{cases} 1, x < x_{min} \\ 0, x \geq x_{min} \end{cases}$$

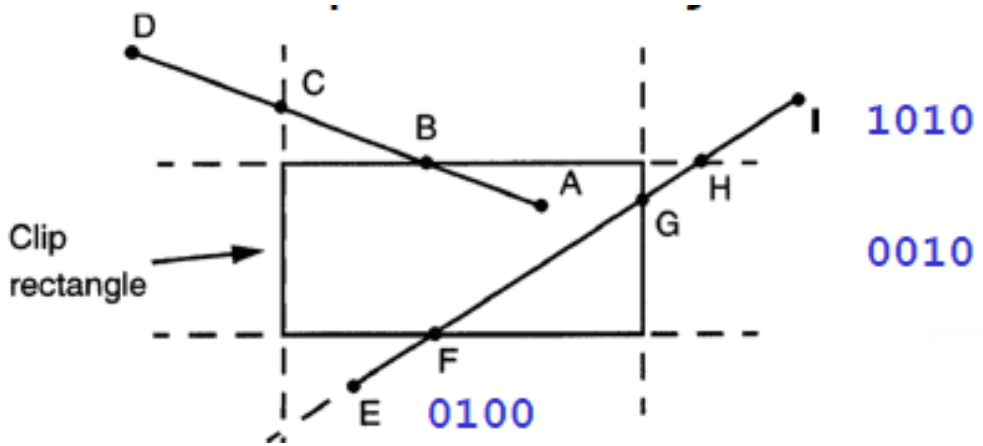
- Let C_0 and C_1 be the TBRL clip code for P_0 and P_1, respectively. Then
  1. If $C_0 = 0$ and $C_1 = 0$ the line is completely inside the domain;
  2. If C_0 and C_1 have one or more bits in common, the line is completely outside the domain;
  3. If any of the bits in C_0 or C_1 are set, clip to the corresponding edge as follows, updating the end points and clip codes as appropriate

$T : x = x_0 + (y_{max} - y_0).\frac{x_1 - x_0}{y_1 - y_0}$ $B : x = x_0 + (y_{min} - y_0).\frac{x_1 - x_0}{y_1 - y_0}$

$R : y = y_0 + (x_{max} - x_0).\frac{y_1 - y_0}{x_1 - x_0}$ $L : y = y_0 + (x_{min} - x_0).\frac{y_1 - y_0}{x_1 - x_0}$

Cohen Sutherland

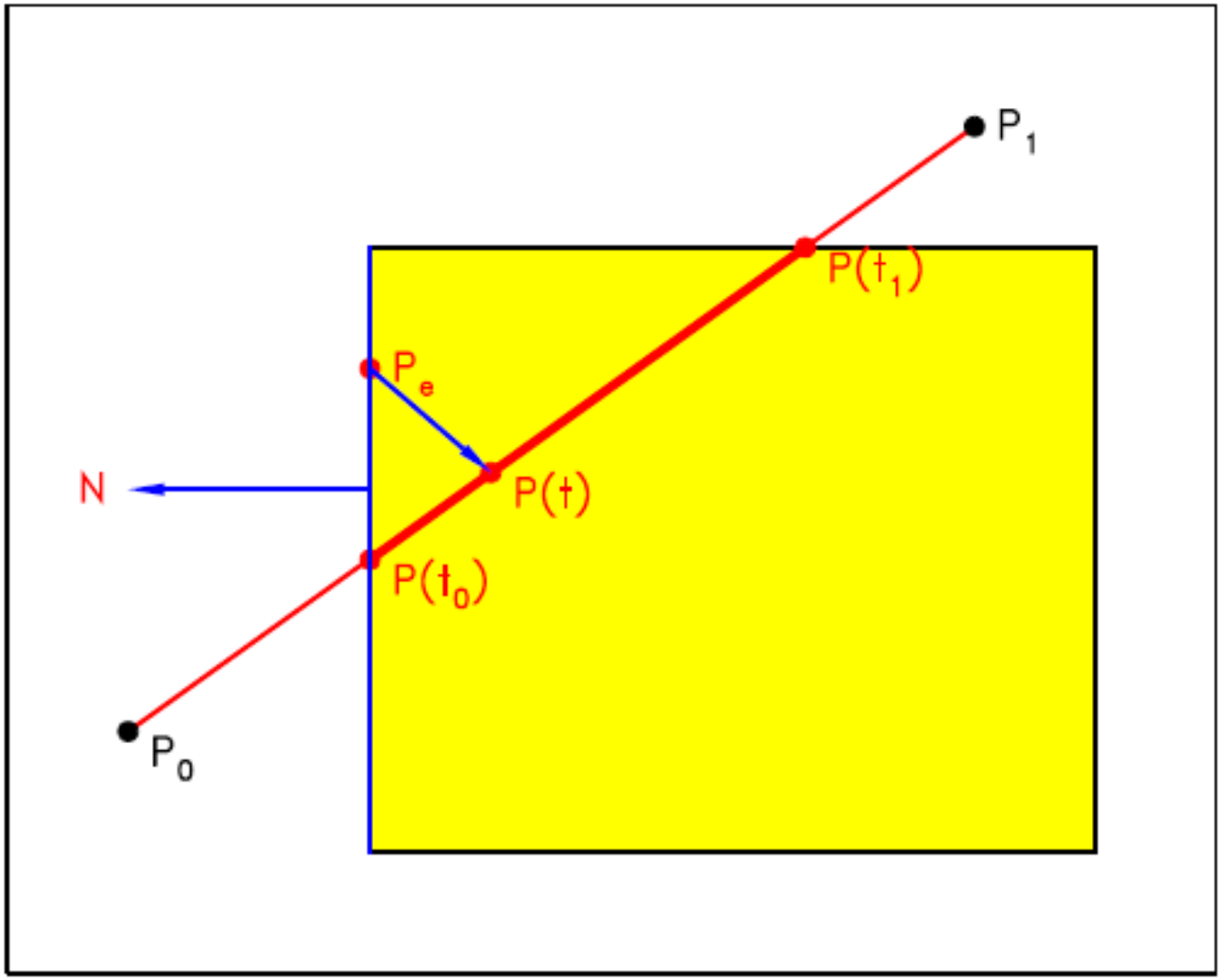
- Select a vertex outside
- rank window edges
- divide line at intersection with edge of highest priority
- delete line segment
- compute outcode
- iterate until trivial accept or trivial reject



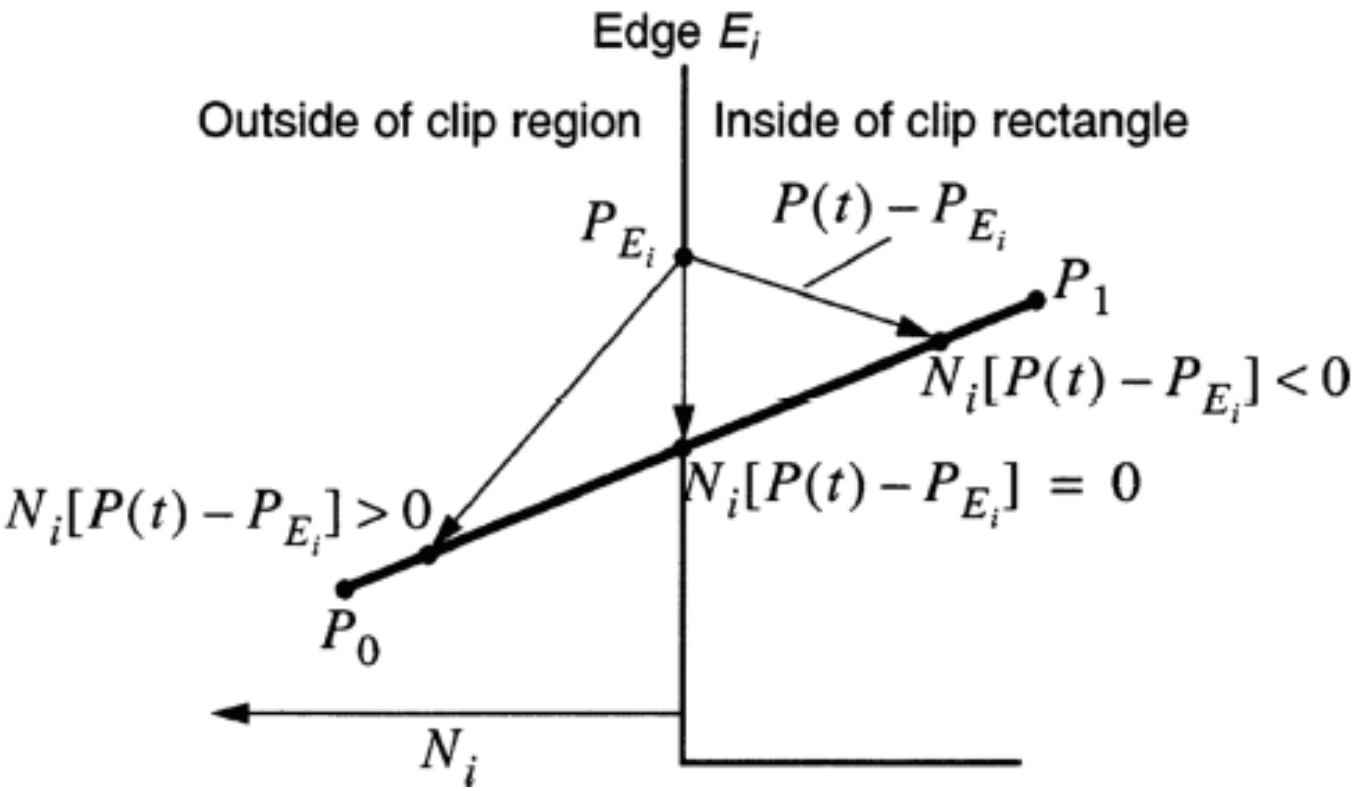
Next

Liang Barsky clipping algorithm

Liang Barsky



Liang Barsky



Liang Barsky

Let $\mathbf{P}(t) = (1-t)\mathbf{P}_0 + t.\mathbf{P}_1$. For $\mathbf{P}(t)$ on the same edge as $\mathbf{P}_e$ and hence perpendicular to $\mathbf{N}$,

$$0 = \mathbf{N}[\mathbf{P}(t) - \mathbf{P}_e]$$

$$= \mathbf{N}.[(1-t).\mathbf{P}_0 + t.\mathbf{P}_1 - \mathbf{P}_e]$$

$$= \mathbf{N}.\mathbf{P}_0 - t.\mathbf{N}.\mathbf{P}_0 + t.\mathbf{N}.\mathbf{P}_1 - \mathbf{N}.\mathbf{P}_e$$

Bring all terms containing t to the left,

$$t.\mathbf{N}.[\mathbf{P}_0 - \mathbf{P}_1] = \mathbf{N}.[\mathbf{P}_0 - \mathbf{P}_e]$$

Rearrange

$$t = \frac{\mathbf{N}.[\mathbf{P}_0 - \mathbf{P}_e]}{\mathbf{N}.[\mathbf{P}_0 - \mathbf{P}_1]}$$

$$t = \frac{\mathbf{N}.[\mathbf{P}_0 - \mathbf{P}_e]}{\mathbf{N}.[\mathbf{P}_0 - \mathbf{P}_1]}$$

For the left edge $\mathbf{N} = (-1, 0)$ and $\mathbf{P}_e = (x_{min}, y)$ so that

$$t = \frac{(-1, 0).(x_0 - x_{min}, y_0 - y)}{(-1, 0).(x_0 - x_1, y_0 - y_1)} = \frac{x_{min} - x_0}{x_1 - x_0}$$

For the right edge $\mathbf{N} = (1, 0)$ and $\mathbf{P}_e = (x_{max}, y)$ so that

$$t = \frac{(1, 0).(x_0 - x_{max}, y_0 - y)}{(1, 0).(x_0 - x_1, y_0 - y_1)} = \frac{x_0 - x_{max}}{x_0 - x_1}$$

$$t = \frac{\mathbf{N}.[\mathbf{P}_0 - \mathbf{P}_e]}{\mathbf{N}.[\mathbf{P}_0 - \mathbf{P}_1]}$$

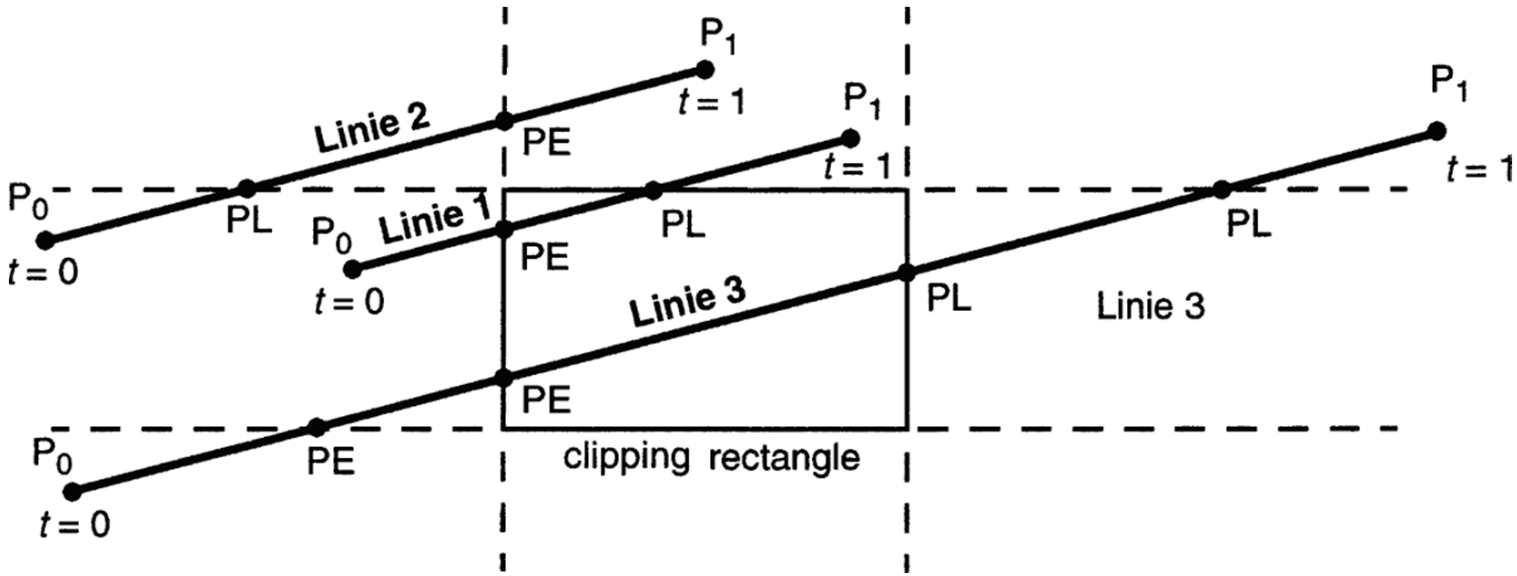
For the bottom edge $\mathbf{N} = (0, -1)$ and $\mathbf{P}_e = (x, y_{min})$ so that

$$t = \frac{(0, -1).(x_0 - x, y_0 - y_{min})}{(0, -1).(x_0 - x_1, y_0 - y_1)} = \frac{y_{min} - y_0}{y_1 - y_0}$$

For the right edge $\mathbf{N} = (0, 1)$ and $\mathbf{P}_e = (x, y_{max})$ so that

$$t = \frac{(0, 1).(x_0 - x, y_0 - y_{max})}{(0, 1).(x_0 - x_1, y_0 - y_1)} = \frac{y_0 - y_{max}}{y_0 - y_1}$$

Liang Barsky Entry and exit criteria



Entry and exit points can be found, Let $\mathbf{D} = \mathbf{P}_1 - \mathbf{P}_0$

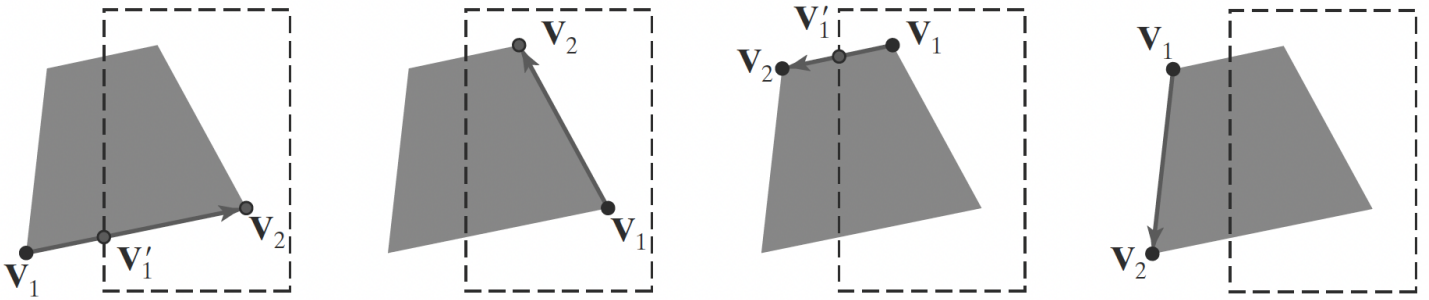
- test for all $t \in [0, 1]$
- $\mathbf{N}.\mathbf{D} < 0 \Rightarrow$ entry $\mathbf{N}.\mathbf{D} > 0 \Rightarrow$ exit.
- Compute maximum (entry) $t_e = max(t_{ei})$ and minimum (exit) $t_l = min(t_{li})$.
- if $t_e > t_l$ no intersection.

Polygon clipping

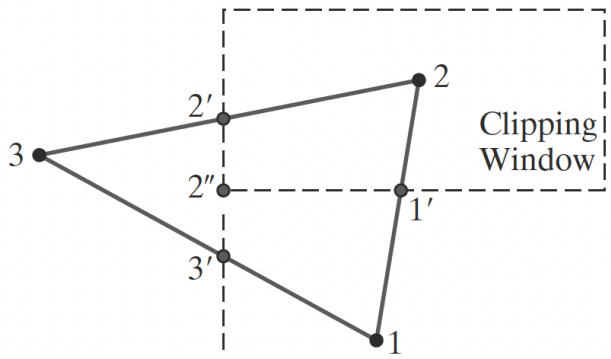
Same but for polygons.
Two algorithms

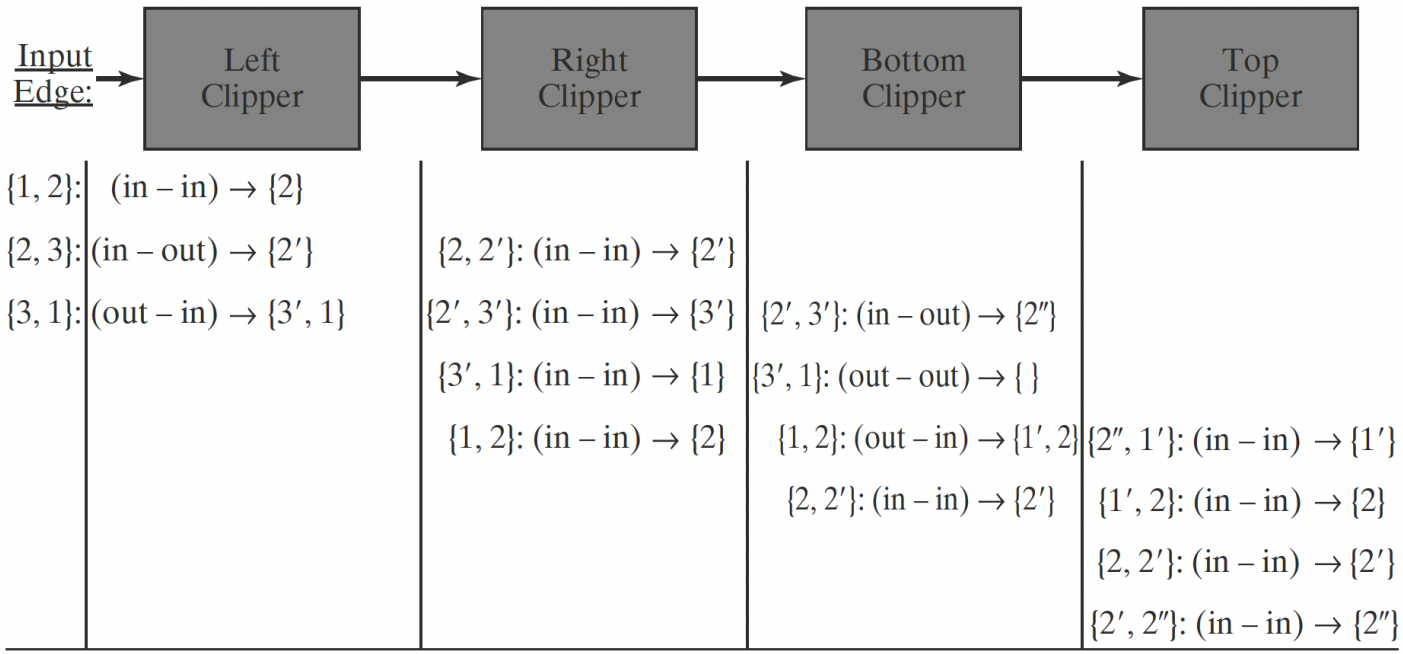
1. Sutherland-Hodgeman algorithm
2. Weiler-Atherton algorithm

Sutherland-Hodgeman algorithm

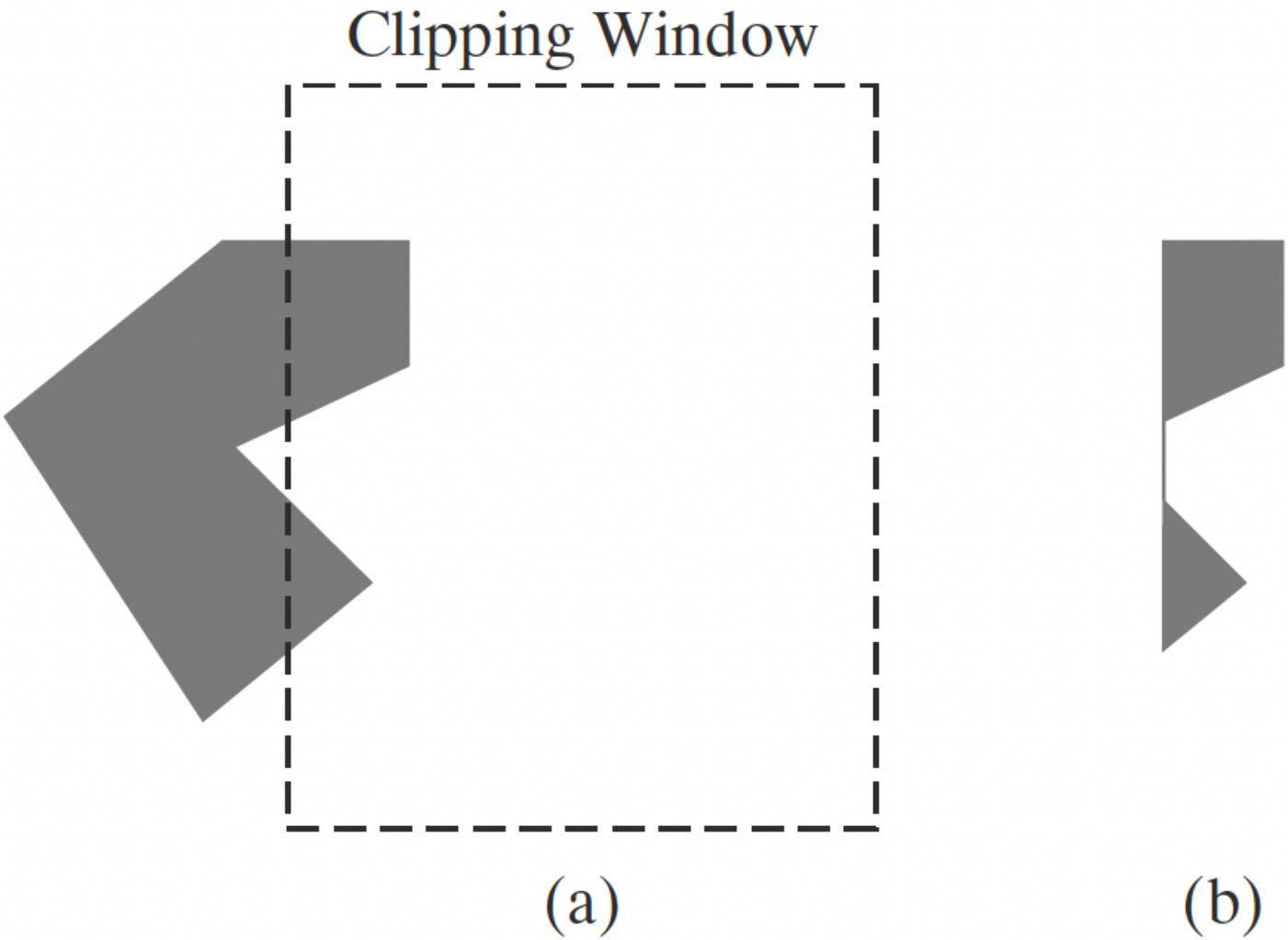


Sutherland-Hodgeman algorithm example

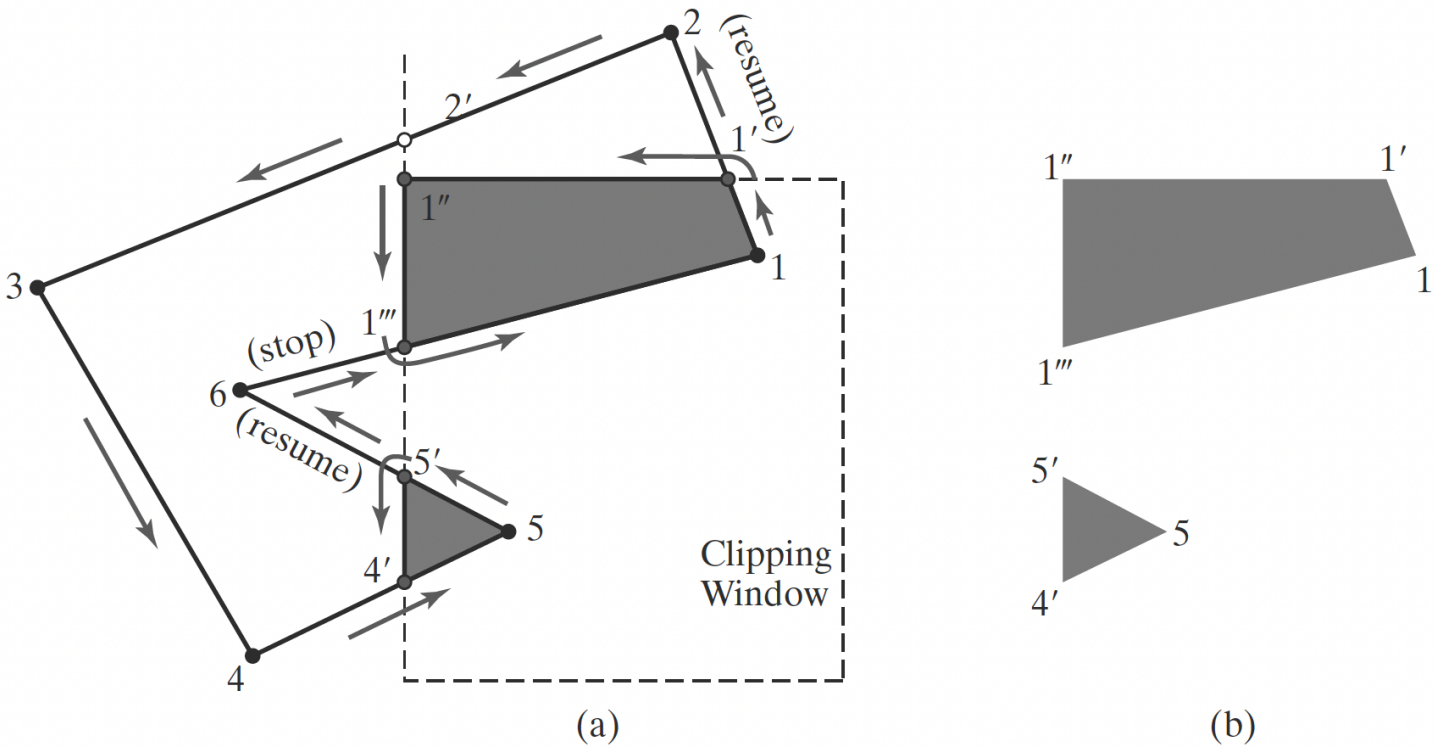




Exception



Weiler-Atherton algorithm



The End